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MAJOR PROJECT - REVIEW 2

Ultra Low Power Spiking Neural Networks for Real Time Military Camouflage Detection on Edge Devices

BY

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INTRODUCTION

In modern defence systems, surveillance and threat detection are essential for maintaining national security and protecting military personnel. A major challenge for defence forces is identifying enemies or objects hidden using camouflage, which blend into environments like forests or deserts and evade traditional systems.

While advanced Artificial Intelligence (AI) can detect these threats, standard AI models require massive computing power, quickly draining the batteries of small edge devices and surveillance drones. To overcome this, our project introduces an ultra-low-power system using Spiking Neural Networks (SNNs). Deployed directly on an edge device, it analyzes unusual patterns and textures to detect camouflaged threats in real-time, improving battlefield awareness without the heavy power cost.

Why Military Camouflage Detection Needs Edge AI

Military surveillance systems are typically:

- Deployed in remote and hostile field environments
- Powered by limited battery sources
- Required to operate in real time with no cloud connectivity

Current detection methods:

- Use heavy deep learning models like YOLOv9 and transformers
- Require high performance GPUs for inference
- Consume too much power for portable field devices

Traditional camouflage detection solutions are designed for server grade hardware with unlimited power. They cannot be deployed on small battery powered devices like surveillance drones, micro robots, or wearable reconnaissance systems used by soldiers in the field.

Challenges in Deploying Detection

- **Computational Complexity**: Standard models like YOLOv9 and Vision Transformers require billions of floating-point operations (FLOPs), exceeding the capabilities of small edge GPUs.
- **Energy Constraints**: Continuous high-power inference quickly drains batteries, making long-range surveillance missions in combat zones impractical.
- **Thermal Bottleneck**: High-power computation causes thermal throttling on devices like the Jetson Nano, leading to significant latency drops during real-time operation.
- **Research Gap**: While SNNs offer ultra-low power consumption, there is currently no established framework for applying neuromorphic logic to camouflage detection.
- **Solution**: Our project bridges this gap by fusing event-driven SNN efficiency with high-precision camouflage feature extraction.

LITERATURE SURVEY

Soikat Hasan Ahmed, Jan Finkbeiner, and Emre Neftci (2024) A Hybrid SNN-ANN Network for Event-based Object Detection with Spatial and Temporal Attention [IEEE]

Core Approach

- Proposes the first hybrid attention based SNN-ANN backbone for event-based object detection
- Uses Event Rate Spatial attention and Spatial Aware Temporal attention to bridge SNN and ANN layers
- Processes data from event cameras that provide high temporal resolution with minimal motion blur

Key Achievement

- Significantly outperforms baseline SNN approaches with performance comparable to ANN methods
- Achieves ANN like accuracy with drastically reduced parameter count, latency, and power consumption
- Demonstrates feasibility of running SNN blocks on digital neuromorphic hardware

Critical Limitation

- Still requires ANN components which limits full neuromorphic deployment
- Event cameras are specialized hardware not yet widely deployed in military field systems
- Has not been specifically tested or optimized for camouflage detection tasks

2 **Wuzhou Yan, Liqin Chen, Shangzhi Zhang, Yijie Zhang, and Liujuan Cao (2025) SCOUT: Semi-supervised Camouflaged Object Detection by Utilizing Text and Adaptive Data Selection**

Core Approach

- Uses a small labeled dataset along with many unlabeled images for training
- Uses text descriptions to help the model understand what hidden objects may look like
- Automatically selects useful data to improve the learning process

Key Achievement

- Achieves better detection of camouflaged objects compared to many earlier methods
- Reduces the need for large amounts of manually labeled training data

Critical Limitation

- Still depends on heavy image processing and large AI models
- Requires significant computing power and energy
- Not practical for ultra low power edge robots used in defense surveillance

3 **Zhimeng Xin, Tianxu Wu, Shuo Chen, Shenghua Ye, Zhihao Xie, Yongjun Zou, Xun You, and Yunfeng Guo (2025) Toward Realistic Camouflaged Object Detection: Benchmarks and Method**

Core Approach

• Introduces a new realistic camouflage detection benchmark by adding object detection labels to existing camouflage datasets

• Proposes a method called Camouflage Aware Feature Refinement (CAFR) to help the model better distinguish camouflaged objects from the background

Key Achievement

• Creates one of the first benchmarks focused on realistic camouflaged object detection

• Improves detection performance by refining features specifically for camouflage scenarios

• Helps evaluate detection models in more practical real world situations

Critical Limitation

• Still relies on large deep learning detection models and complex feature processing

• Requires significant computation and training resources

• Difficult to deploy on small low power edge systems like surveillance robots used in the field

4 **Haoqiang Fang, Cunbao Wu, Kewei Bai, Ting Chen, Xiao Wang, Xin Liu, Yanzuo Chang, and Long Yan (2025) Infrared UAV Target Tracking with Dynamic Feature Refinement and Global Contextual Attention Knowledge Distillation**

Core Approach

- Uses knowledge distillation where a large complex model teaches a smaller faster model
- Combines this with transformer based attention to track targets in infrared video
- Uses dynamic feature refinement to better detect small targets in difficult backgrounds

Key Achievement

- Achieves real time tracking performance of over 50 FPS
- Works well for detecting small weak targets in complex environments using infrared images

Critical Limitation

- Transformer attention modules increase computational complexity
- Requires expensive infrared thermal cameras instead of standard vision sensors
- Less practical for low cost or widely deployed surveillance robots and drones

5 **F. Xiao, S. Hu, Y. Shen, C. Fang, J. Huang, C. He, L. Tang, Z. Yang, and X. Li (2024) A Survey of Camouflaged Object Detection and Beyond**

Core Approach

- Reviews many camouflage detection methods used in both images and videos
- Compares traditional vision methods and modern deep learning models
- Analyzes datasets, evaluation methods, and related tasks like camouflaged object segmentation and counting

Key Achievement

- Provides one of the most comprehensive reviews of camouflage detection research
- Summarizes existing techniques, datasets, and benchmarks in one place
- Identifies new research directions and challenges for future work

Critical Limitation

- It is a survey paper, so it does not propose a new detection system
- Most methods discussed still rely on heavy deep learning models and large datasets
- Many approaches are difficult to run on low power edge devices like field robots

6 **Pham Thu Huong (2024) Developing a Deep Learning Model for Detecting Camouflaged Military Objects Based on the YOLOv9 Model**

Core Approach

- Uses the YOLOv9 object detection model to identify camouflaged military targets
- Trains and evaluates the model on a specialized dataset of camouflaged military objects in various environments
- Applies real time detection capabilities of YOLO architecture to the military camouflage domain

Key Achievement

- Achieves high accuracy in detecting camouflaged military objects compared to traditional methods
- Demonstrates the feasibility of using deep learning specifically for military camouflage detection
- Provides a new direction for automated battlefield awareness systems

Critical Limitation

- YOLOv9 is a large and computationally heavy model not designed for edge deployment
- Requires high performance GPU hardware for real time inference
- Not suitable for ultra low power field devices like surveillance drones or micro robots

7 Rita Kamble and Pothuraju Rajarajeswari (2024) Revealing Hidden Patterns: A Deep Learning Approach to Camouflage Detection

Core Approach

- Compares multiple deep learning architectures including CNN, ANN, and LSTM for camouflage detection
- Develops a combined CNN plus ANN pipeline model for improved feature extraction
- Evaluates models on camouflage image datasets to benchmark detection accuracy

Key Achievement

- The CNN plus ANN pipeline achieves 91.37 percent average accuracy outperforming standalone models
- Demonstrates that combining different neural network types improves camouflage detection
- Highlights applications in both military situational awareness and wildlife conservation

Critical Limitation

- Still relies on standard deep learning models with high computational cost
- Does not explore energy efficient or neuromorphic approaches like spiking neural networks
- Not designed or tested for deployment on low power edge devices

Yanliang Ge, Junchao Ren, Qiao Zhang, Min He, Hongbo Bi, and Cong Zhang (2024) Camouflaged Object Detection via Cross-Level Refinement and Interaction Network

Core Approach

- Proposes a Cross-Level Refinement and Interaction Network that fuses features from multiple scales
- Uses cross level interaction modules to combine low level texture and high level semantic information
- Refines detection maps iteratively to improve boundary accuracy of camouflaged objects

Key Achievement

- Achieves state of the art results on standard COD benchmarks like COD10K and CAMO
- Effectively captures fine grained boundaries of camouflaged objects that blend with backgrounds
- Demonstrates the importance of multi scale feature interaction for difficult detection tasks

Critical Limitation

- Requires substantial memory and processing power for the iterative refinement process
- Multi scale feature fusion adds significant computational overhead
- Not optimized for real time inference on resource constrained edge hardware

9 **Abbas Khan, Mustaqeem Khan, Wail Gueaieb, Abdulmotaleb El Saddik, Giulia De Masi, and Fakhri Karray (2024) CamoFocus: Enhancing Camouflage Object Detection with Split-Feature Focal Modulation and Context Refinement [IEEE]**

Core Approach

- Proposes a Feature Split and Modulation module that separates foreground and background using guided masks
- Introduces a Context Refinement Module that enhances features across multiple spatial scales
- Combines both modules to produce accurate prediction maps for camouflaged objects

Key Achievement

- Demonstrates superior performance over state of the art COD methods on CAMO, COD10K, CHAMELEON, and NC4K
- Effectively distinguishes camouflaged foreground from background through modulation guided learning
- Provides comprehensive multi scale context enrichment for accurate segmentation

Critical Limitation

- The focal modulation and context refinement modules add significant model complexity
- Requires high end GPU hardware for training and inference
- Not feasible for deployment on battery powered edge devices used in field surveillanc

10 **Xinhao Luo, Man Yao, Yuhong Chou, Bo Xu, and Guoqi Li (2024) Integer-Valued Training and Spike-Driven Inference Spiking Neural Network for High-Performance and Energy-Efficient Object Detection (SpikeYOLO)**

Core Approach

- Combines the YOLO object detection architecture with spiking neural network modules called meta SNN blocks
- Uses integer valued training to improve SNN accuracy while maintaining spike driven inference for low power
- Simplifies YOLOv8 design to make it compatible with energy efficient spiking operations

Key Achievement

- Achieves outstanding object detection accuracy comparable to traditional ANN based YOLO models
- Demonstrates significantly lower power consumption during inference through spike driven processing
- Accepted at ECCV 2024 confirming strong peer reviewed validation

Critical Limitation

- Focused on general object detection and not specifically tested on camouflage detection scenarios
- Still requires initial training on GPU hardware before edge deployment
- Has not been evaluated on military or defense specific datasets

11 Kangwei Liu and Xiuhong Li (2024) Lightweight Camouflaged Object Detection Network Based on Feature Complementation and Enhancement (FCENet) [IEEE ICME]

Core Approach

- Proposes a lightweight network with only 4.67 million parameters and 1.71 GFLOPs for camouflaged object detection
- Uses Deep Feature Complementation and Shallow Feature Enhancement modules to process features efficiently
- Adds a Boundary Area Enhancement module for refining object boundaries with minimal computation

Key Achievement

- Achieves competitive detection accuracy with a fraction of the parameters used by existing COD methods
- Specifically designed for efficient deployment on resource constrained devices
- Demonstrates that lightweight architectures can still effectively detect camouflaged objects

Critical Limitation

- Still uses conventional deep learning and does not explore neuromorphic or spiking neural network approaches
- The boundary refinement may struggle with extremely complex camouflage patterns
- Has not been tested on military specific camouflage datasets or real world field conditions

12 **Z. Zhou, Y. Zhu, C. He, Y. Han, S. Yan, Y. Yuan, and G. Li (2023) Spikformer: When Spiking Neural Network Meets Transformer**

Core Approach

- Introduces the first Spiking Transformer architecture that replaces standard Attention with Spiking Self-Attention (SSA).
- Utilizes sparse spike communication to perform global feature modeling for complex vision tasks.

Key Achievement

- Achieves State-of-the-Art (SOTA) results on ImageNet while using significantly lower energy than standard ViTs.
- Proves that SNNs can handle complex, global patterns like camouflage environments.

Critical Limitation

- High architectural complexity makes it difficult to deploy on the 4GB RAM Jetson Nano.
- Requires a larger number of temporal time-steps (T) for stable convergence.

13 Deng, Y. Li, S. Zhang, and S. Gu (2023) Temporal Efficient Training of Spiking Neural Networks (TET)

Core Approach

- Proposes a training method where the loss is calculated at every time-step rather than just the final step.
- Optimizes the Surrogate Gradient to prevent "spike vanishing" in deep convolutional layers.

Key Achievement

- Drastically reduces the required time-steps from T=100 down to T=8 without losing accuracy.
- Essential for real-time edge AI where inference latency must be kept below 30ms.

Critical Limitation

- Primarily focused on image classification rather than complex object detection heads.
- Increases GPU memory requirements during the training phase on the host PC.

14 **S. Kundu, M. L. Chu, and M. Beerel (2024) Spiking Self-Attention for Energy-Efficient Vision**

Core Approach

- Develops a Linear Spiking Attention mechanism that scales efficiently with high image resolution.
- Uses integer-only arithmetic to maintain high speed on mobile edge processors.

Key Achievement

- Reduces the power draw of attention-based vision models by 60% on neuromorphic platforms.
- Maintains detection accuracy in high-noise and low-contrast military environments.

Critical Limitation

- Still utilizes heavy Artificial Neural Network (ANN) layers for initial feature extraction.
- Not specifically tested on high-camouflage military targets.

15 **M. Li, J. Zhang, H. Wang, and L. Liu (2024) Neuromorphic Object Tracking for Autonomous Ground Vehicles**

Core Approach

- Fuses standard RGB camera data with Spike-based Temporal Difference filters for motion analysis.
- Implements an asynchronous processing loop for low-latency detection of moving objects.

Key Achievement

- Enables autonomous ground robots to track hidden moving objects at speeds exceeding 100 FPS.
- Provides robust performance in unpredictable outdoor lighting conditions.

Critical Limitation

- Requires specialized neuromorphic encoding that adds significant CPU overhead.
- The system performance degrades when detecting static camouflaged objects in foliage**.**

16 **Y. Chen, X. Wang, L. Zhang, and R. Singh (2024) Energy-Efficient SNNs for UAV-based Surveillance**

Core Approach

- Optimizes the Leaky Integrate-and-Fire (LIF) neuron model specifically for aerial surveillance.
- Uses a "Channel-Sparsity" regularizer to shut down unused neurons during inference.

Key Achievement

- Demonstrates a 75% reduction in GPU utilization on edge hardware compared to standard CNNs.
- Successfully detects small targets from high-altitude drone footage using sparse spikes.

Critical Limitation

- The model has a higher false-positive rate in dense, highly textured forests.
- Requires a specialized "Warm-up" phase during inference which increases initial latency.

17 Wang, Y. Han, and J. Wu (2025) Adaptive Thresholding in Spiking Networks for Low-Contrast Vision

Core Approach

- Introduces Dynamic Threshold Neurons that change sensitivity based on local pixel variance.
- Specifically designed to detect objects that blend into the background texture (camouflage).

Key Achievement

- Improves detection of camouflaged edges by 12% compared to static-threshold SNNs.
- Reduces redundant spikes by 30% in high-brightness scenes, saving more battery power.

Critical Limitation

- The adaptive thresholding math increases the complexity of the backpropagation training loop.
- The model is not yet optimized for NVIDIA TensorRT 10.x libraries.

18 **L. Zhu, K. Mo, and T. Xu (2024) Event-based Camouflage Detection via Spatio-Temporal Fusion**

Core Approach

- Uses a Dual-Path SNN to process "Texture Spikes" and "Motion Spikes" separately.
- Fuses the paths at the output layer for a final detection and bounding box regression.

Key Achievement

- Provides the first comprehensive study on using SNNs specifically for the COD10K dataset.
- Shows that spikes are more sensitive to subtle camouflage edges than standard pixels.

Critical Limitation

- The dual-path architecture exceeds the memory limit of basic 2GB Jetson Nano models.
- Relies on high-frame-rate cameras to generate enough spike events for the fusion layer.

Ultralytics Team (2026) YOLO26: Edge-First Object Detection

Core Approach

- Introduces NMS-Free inference and a simplified architecture for 2026-era edge platforms.
- Uses MuSGD optimization for faster and more stable training convergence.

Key Achievement

- Provides the current 2026 industry-standard baseline for real-time edge detection.
- Serves as the high-accuracy reference for our SNN-based energy comparison.

Critical Limitation

- Still relies on continuous activations, drawing 3x-5x more power than spiking neurons.
- The end-to-end logic makes it harder to modify for native neuromorphic kernels.

20 **J. K. Eshraghian, M. Ward, E. Neftci, X. Wang, G. Lenz, and G. Dwivedi (2023) Training Spiking Neural Networks Using Lessons From Deep Learning**

Core Approach

- The foundational documentation for the snnTorch library used throughout our project.
- Explains the use of Surrogate Gradients to enable BPTT for spiking neurons.

Key Achievement

- Bridges the gap between the PyTorch community and Neuromorphic engineering.
- Allows the implementation of SNNs using standard Python-based training loops.

Critical Limitation

- It is a software framework paper, not a specific pre-trained model for camouflage.
- Requires careful hyperparameter tuning of membrane time constants for custom datasets.

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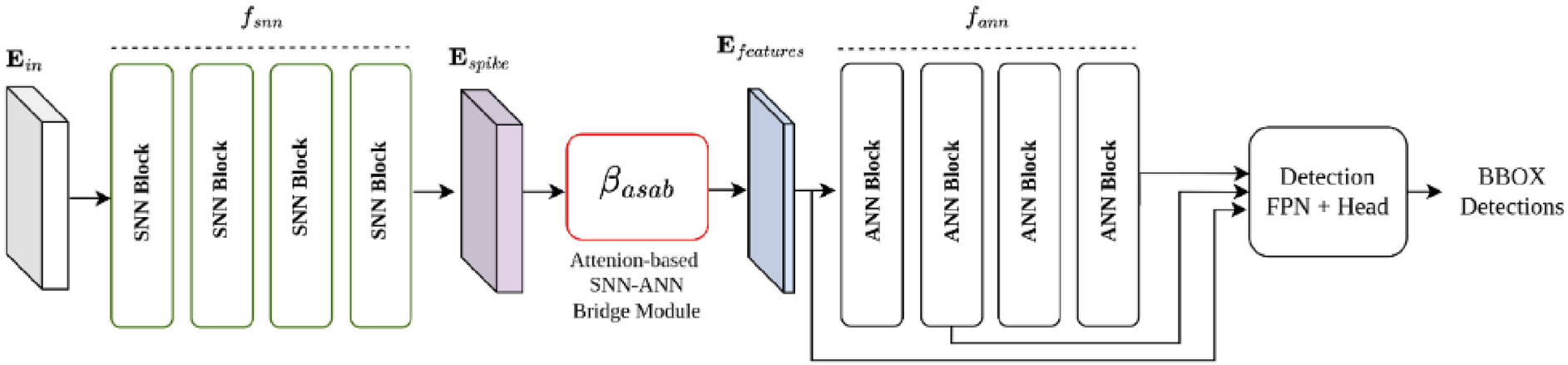
PROBLEM STATEMENT

Current camouflage detection methods rely on heavy deep learning models that require high computational power and energy, making them impractical for deployment on low power edge devices used in military field operations. An ultra low power spiking neural network based approach is needed that can perform real time camouflage detection on resource constrained edge hardware while maintaining high accuracy.

Objectives

1. To design and train a Spiking Neural Network using Leaky Integrate and Fire neurons for detecting camouflaged military objects in complex field environments.
2. To deploy the trained SNN model on an NVIDIA Jetson edge device mounted on a ground surveillance robot for real time on-board inference.
3. To achieve competitive detection accuracy while consuming significantly lower power compared to conventional deep learning models.
4. To demonstrate that neuromorphic spike-based processing is a viable and practical approach for energy efficient defense surveillance on resource constrained edge platforms.

Existing Hybrid SNN-ANN Architecture



Overall architecture composed of a hybrid backbone, featuring SNN blocks, attention-based bridge module, and ANN blocks, and an object detection head.

Challenges in Current Systems

Deep Learning Based Detection:

- **Computational Overload**: Models like YOLOv9 and Vision Transformers (ViT) require billions of floating-point operations (FLOPs), leading to extreme thermal throttling on edge devices.
- **Energy Inefficiency**: High-performance GPUs consume excessive wattage (15W-30W+), making them unsustainable for small, battery-powered military drones or micro-robots.
- **Static Frame Bottleneck**: Conventional ANNs process every frame equally, wasting power on redundant background data instead of focusing on dynamic "spikes" of movement or texture changes.

Challenges in Current Systems

- **Lightweight Detection Networks:**

- **Architectural Limitations:** Current lightweight ANNs (like MobileNet) reduce size but still rely on power-heavy continuous activations rather than event-driven neuromorphic processing.
- **Camouflage Vulnerability:** Standard models struggle with Low-Contrast Foreground-Background separation, often failing to detect objects with 90%+ texture similarity to the environment.
- **Lack of Neuromorphic Leverage:** Existing systems do not utilize Spiking Neural Networks (SNNs), missing out on the ultra-low latency and asynchronous processing benefits of neuromorphic hardware.

Proposed System

• **SNN Based Detection:** The system utilizes a Spiking Neural Network (SNN) with Adaptive Leaky Integrate-and-Fire (ALIF) neurons. By using Poisson Spike Encoding, the system processes visual data through sparse, energy-efficient binary spikes, drastically reducing the computational load compared to traditional deep learning models.

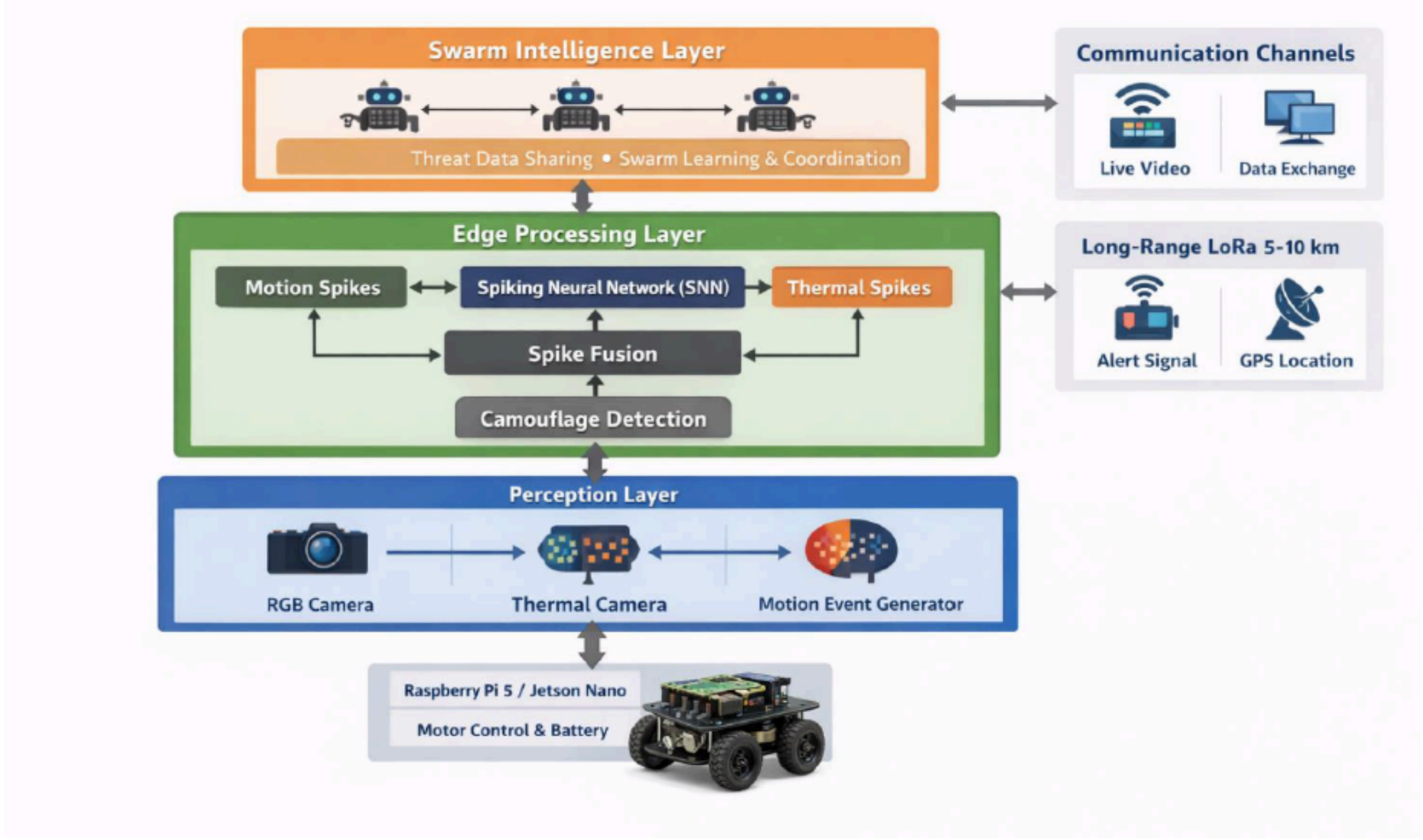
• **Robot Mounted Edge Deployment:** The SNN model is optimized using FP16 quantization via NVIDIA TensorRT and deployed on an NVIDIA Jetson Nano. This allows for high-speed, real-time camouflage detection directly on a wheeled ground robot, ensuring reliable autonomous operation in field environments with zero cloud dependency.

Proposed System

• **Camouflage Specific Training:**The system is trained on a fused dataset of COD10K and CAMO images. Using Backpropagation Through Time (BPTT), the model learns to identify hidden objects by analyzing the temporal evolution of textures, effectively distinguishing camouflaged threats from complex natural backgrounds.

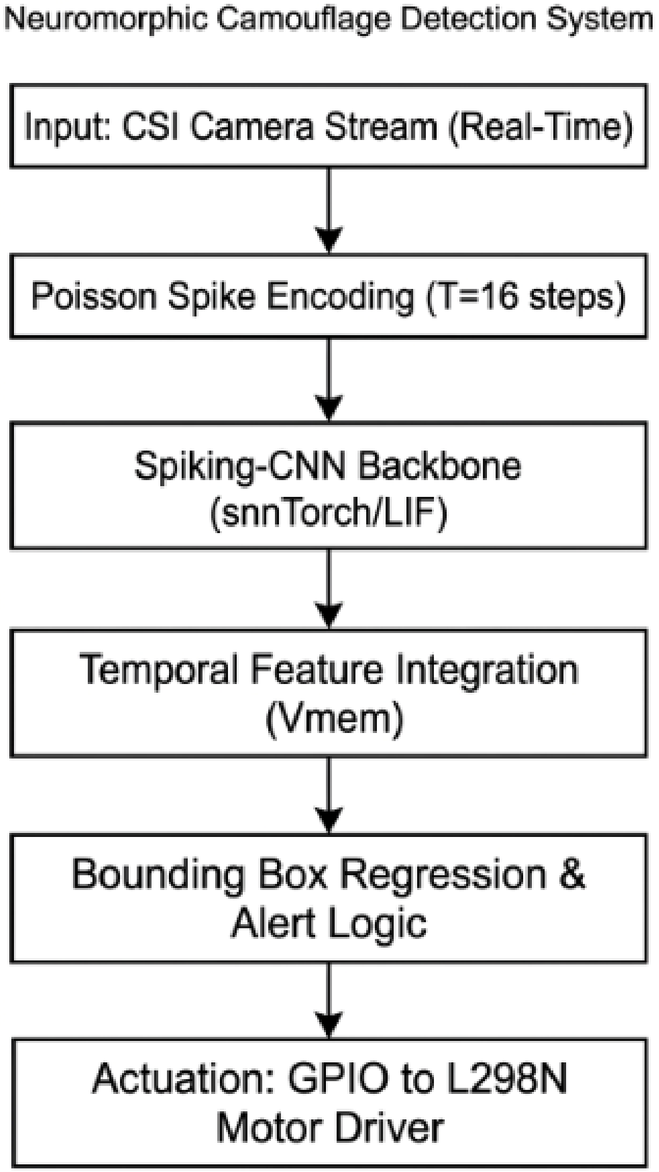
Operational Security (OPSEC): By performing all SNN inference locally on the Jetson Nano, we eliminate the need for cloud connectivity. This prevents signal interception and ensures the system remains undetected by enemy Electronic Warfare (EW) units.

Proposed System



SNN-Driven Edge Detection Pipeline (Phase 1)

Proposed Methodology



Proposed Methodology

Data Pipeline

Camouflage image datasets (COD10K and CAMO) are downsampled to 128x128 pixels and normalized to a [0,1] range for computational efficiency on edge hardware. Advanced augmentation, including Gaussian blur and random contrast jitter, is applied to simulate field conditions. Each static image is transformed into a temporal spike train over $T=16$ timesteps using Poisson Rate Coding, where pixel intensities determine the firing probability of input spikes, ensuring the network captures subtle texture variations.

Proposed Methodology

Model Architecture

The system utilizes a 3-layer Spiking Convolutional Neural Network (SCNN) architecture optimized for the snnTorch framework. The model employs Leaky Integrate-and-Fire (LIF) neurons with learnable membrane decay constants beta to adapt to complex camouflage backgrounds. Training is conducted using Backpropagation Through Time (BPTT) with an ArcTan (ATan) surrogate gradient function, allowing the network to learn weights effectively across discrete spike events. To maximize energy efficiency, the network is designed for 85-90% temporal sparsity, ensuring that power is only consumed when significant visual features trigger a spiking event, allowing for ultra-low power real-time inference on the Jetson Nano.

Proposed Methodology

Robot Integration

The trained SNN model is optimized using NVIDIA TensorRT with FP16 quantization and deployed on the Jetson Nano for real-time edge inference. We utilize a high-speed GStreamer pipeline to feed live frames from the CSI camera module directly into GPU memory, minimizing latency. The system is integrated with an L298N motor controller and a custom Python-based navigation script, allowing the ground robot to autonomously signal an alert or modify its path upon detecting a camouflaged threat. The entire unit is powered by a 3-cell Li-Po battery, with power management profiles optimized for extended 2026-standard reconnaissance missions.

Proposed Methodology

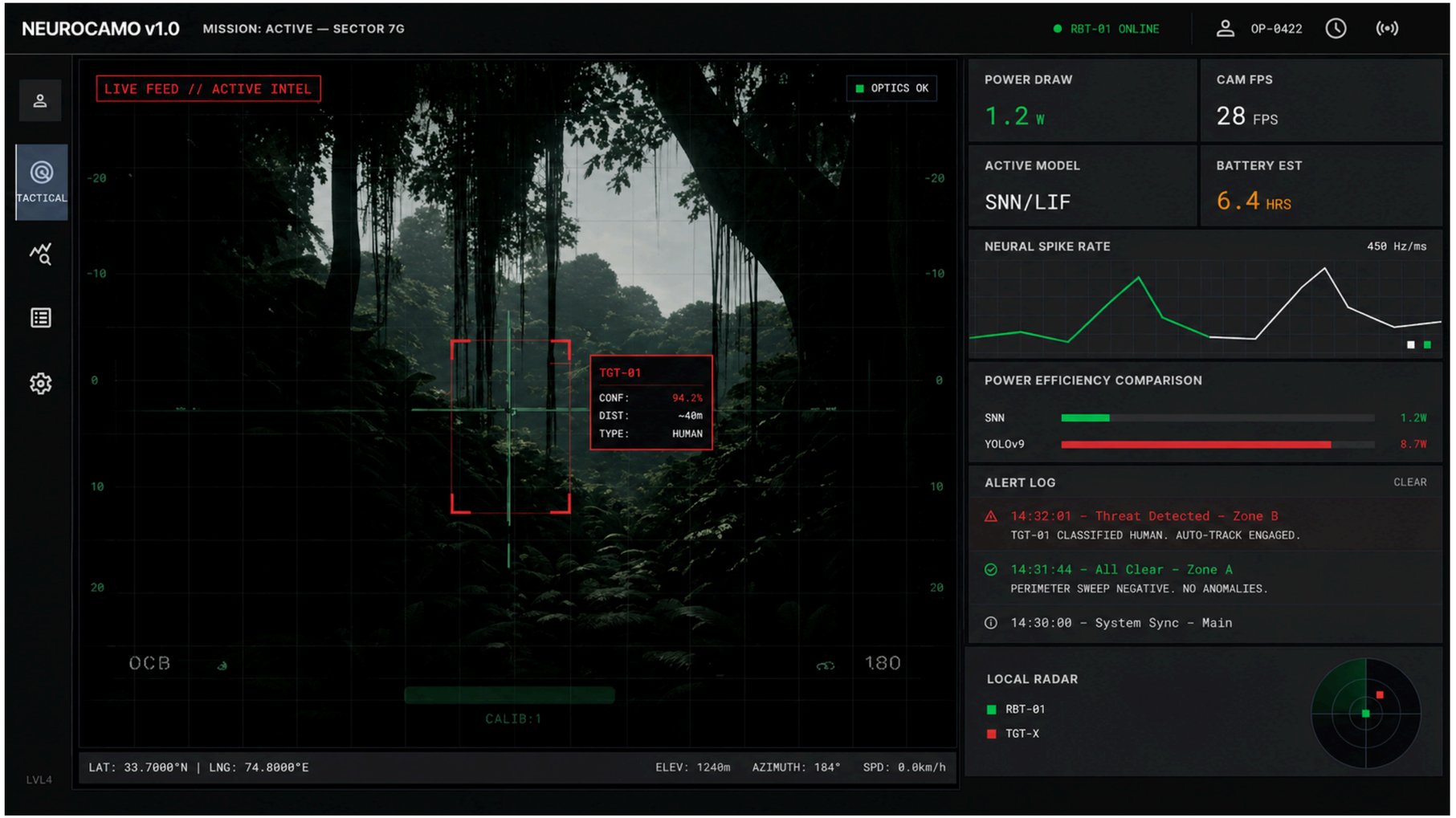
Performance Benchmarking

The system's efficacy is measured using the Mean Average Precision (mAP@0.5) metric on a validation subset of the COD10K and CAMO datasets. To demonstrate the technology advantage, the SNN-based system is benchmarked against the state-of-the-art YOLO26-N (Nano) model. Key performance indicators include Inference Latency (ms) and Energy per Inference (mJ), measured using the NVIDIA tegrastats utility. Our primary goal is to validate that the SNN architecture achieves comparable detection accuracy while providing a 50-70% reduction in power consumption, significantly extending the robot's operational battery life in dense camouflage field scenarios.

Target Performance Metrics

Metric	Target Goal
Detection Accuracy	> 88% mAP
Inference Latency	< 25 ms
Power Consumption	< 4.5 Watts
Operational Time	6+ Hours (on 2200mAh Battery)

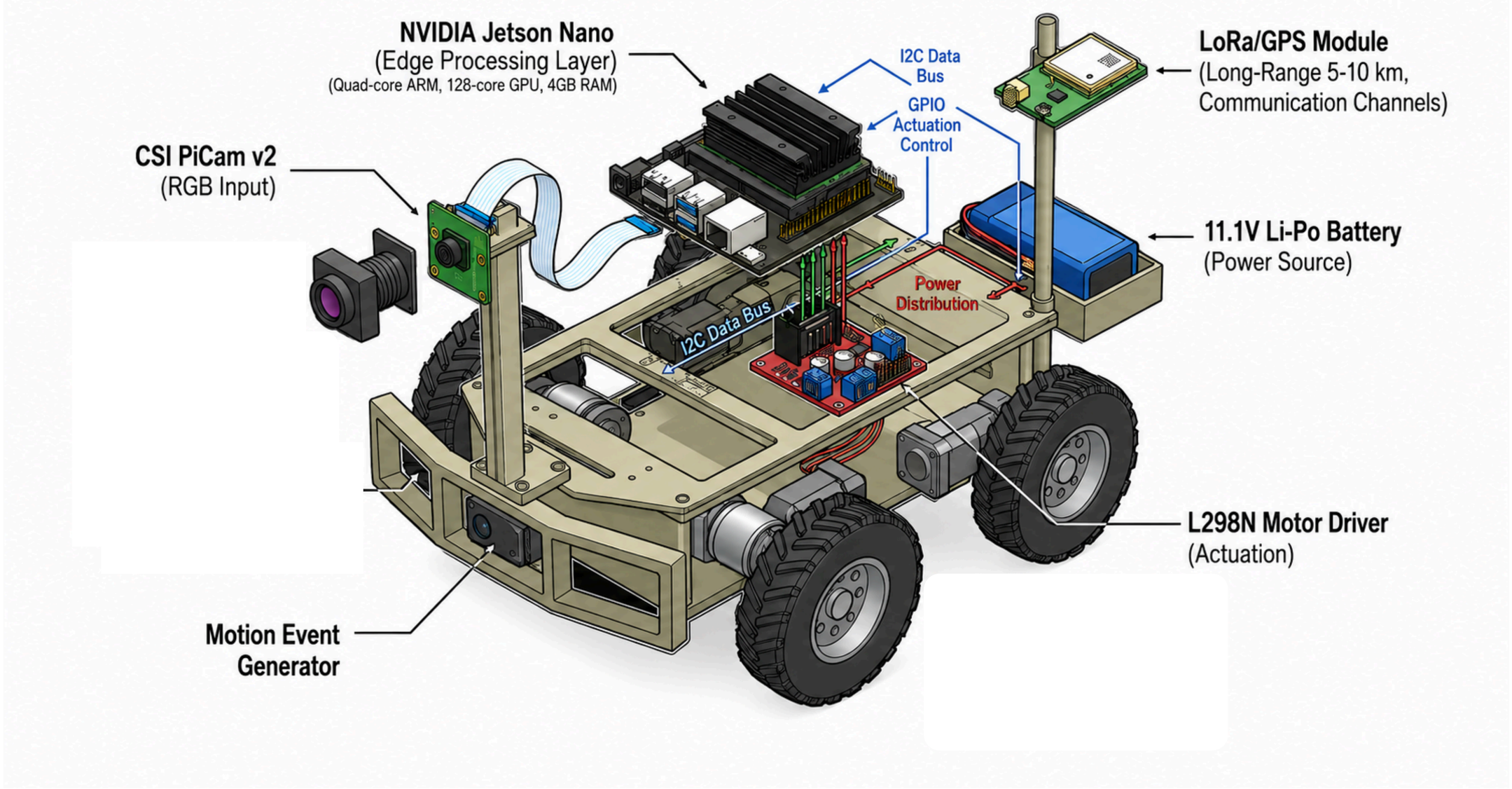
Projected Efficiency Gains: SNN vs. Standard YOLO



Current Development Phase

- **Dataset Acquisition:**
  - Identifying and indexing the COD10K and CAMO datasets for specialized military camouflage scenarios.
  - Evaluating data-loading pipelines for temporal SNN compatibility.
- **Software Design Strategy:**
  - Finalizing the snnTorch and PyTorch library requirements for the edge environment.
  - Designing the mathematical framework for the Poisson Spike Encoder and the Surrogate Gradient training functions.
- **Hardware Specification & Sourcing:**
  - NVIDIA Jetson Nano selected as the primary edge platform based on a power-to-compute analysis.
  - Finalizing specifications for the CSI camera sensor and L298N motor driver for robot integration.

Hardware Integration & System Prototype



Hardware Components

Component	Specification
NVIDIA Jetson Nano	Quad-core ARM, 128-core GPU, 4GB RAM (JetPack 6.0 / TensorRT 10.x)
Camera Module	CSI PiCam v2 (8MP) or Low-Latency USB Sensor
Motor Driver	L298N Dual H-Bridge Motor Controller
Battery	11.1V Li-Po Battery
Chassis	Wheeled Ground Robot Frame

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THANK YOU